

Document	SharePoint Backend Schema
Project	Luxury Retail Employee Flash Sale Operations System
Developer	Nandani Raj
Data Layer	Microsoft SharePoint Online — 8 structured lists
Date	June 2026
Version	1.0

Overview

The SharePoint backend is the data layer for the entire system. All eight lists are connected through a shared SaleID field that acts as a foreign key across the system. The Power App reads from and writes to these lists. Power Automate flows are triggered by changes in these lists. Power BI connects to these lists as its data source.

Before building the Power App or any automation flows, all eight lists must be created with the correct column internal names, data types, and Choice field values. Column internal names in SharePoint may differ from display names — always verify the internal name in List Settings before referencing a column in Power Fx formulas.

SaleID — portfolio vs. production: The SaleID field is populated by a Power Fx formula (Concatenate + CountRows) when a new sale record is created. This approach is acceptable for portfolio demonstration but is not production-safe — concurrent submissions can produce duplicate IDs. In production, use the SharePoint item ID (auto-incremented), GUID(), or a Power Automate-generated sequence.

List Relationships

All lists relate to Sale Master via SaleID. The diagram below shows the relationship structure:

Sale Master (SaleID – primary key) ■■■ Employee Access Requests (SaleID foreign key) ■ ■■■
Approval Log (RequestID + SaleID) ■■■ Issue Tracker (SaleID foreign key) ■■■ Notification
Log (SaleID foreign key) ■■■ Change History (RecordID maps to SaleID) ■■■ Sale Readiness
Checklist (SaleID foreign key) User Role Reference – standalone, linked via UserEmail on app
load

List 1 — Sale Master List

Purpose: The authoritative record of every flash sale event. All other lists reference this list via SaleID. Admins create and manage records here; employees never write to this list directly.

Relationships: SaleID is the primary key referenced as a Lookup or matching text field in all 7 related lists.

Column Name	Data Type	Req	Example Value	Purpose / Notes
SaleID	Single line text	Yes	SALE-2026-001	Unique identifier — foreign key in all related lists. Auto-generated on form submit.
SaleName	Single line text	Yes	Spring 2026 Employee Appreciation Sale	Human-readable name displayed across all app screens.
SaleStatus	Choice	Yes	Active	Core lifecycle field. Values: Draft; Open for Requests; Readiness Review; Active; Closing; Closed; Archived.
SaleStartDate	Date and Time	Yes	5/15/2026 9:00 AM	Used in countdown displays, expiry SQL checks, and scheduled flow triggers.
SaleEndDate	Date and Time	Yes	5/16/2026 6:00 PM	End boundary for active access. Used in SQL Script 03 expiry check.
DiscountPercentage	Number	Yes	40	Displayed on Sale Details screen and access request confirmation.
EligibleDepartments	Multiple lines	Yes	Retail Operations; HR; Finance	Eligibility validation in Access Request screen.
EligibleLocations	Multiple lines	Yes	New York Flagship; Los Angeles	Scopes which location employees may request access.
ProductCategories	Multiple lines	Yes	Handbags; Footwear; Accessories	Displayed on Sale Details so employees know what they can purchase.
MaxParticipants	Number	No	150	Optional ceiling on approved requests — enforced via Power Fx before submit.
LaunchAuthorizedBy	Single line text	No	Amanda Chen	Admin who authorised launch. Populated on status change to Active.
LaunchAuthorizedDate	Date and Time	No	5/14/2026 4:30 PM	Compliance timestamp — records when go-live was approved.
TotalApproved	Number	No	71	Denormalized KPI updated by Post-Sale Summary Flow. Used in Power BI.
TotalRequests	Number	No	84	Denominator for approval rate KPI.

List 2 — Employee Access Request List

Purpose: Stores every employee access request submitted through the app. RequestStatus drives the approval flow trigger and all app filtering logic.

Relationships: SaleID links to Sale Master. RequestID is referenced in Approval Log and Notification Log.

Column Name	Data Type	Req	Example Value	Purpose / Notes
RequestID	Single line text	Yes	REQ-2026-0047	Unique identifier. Referenced in Approval Log and notification emails.

Column Name	Data Type	Req	Example Value	Purpose / Notes
SaleID	Lookup / text	Yes	SALE-2026-001	Links request to specific sale. Drives all filtering in admin screens.
SaleName	Single line text	Yes	Spring 2026 Sale	Cascaded from Sale Master for readability in forms and notifications.
EmployeeName	Single line text	Yes	Sofia Marchetti	Displayed in admin access management and approval emails.
EmployeeEmail	Single line text	Yes	s.marchetti@luxemaison.com	Recipient of approval decision notification emails.
EmployeeDepartment	Choice	Yes	Retail Operations	Validated against SaleMaster.EligibleDepartments before submit.
EmployeeLocation	Choice	Yes	New York Flagship	Validated against SaleMaster.EligibleLocations.
ManagerName	Single line text	Yes	James Whitfield	Shown in admin log and included in approval request email.
ManagerEmail	Single line text	Yes	j.whitfield@luxemaison.com	Recipient of Power Automate approval email (Flow 1).
RequestStatus	Choice	Yes	Approved	Values: Pending; Approved; Denied; Pending Revocation; Revoked. Drives all flow triggers.
RequestDate	Date and Time	Yes	5/10/2026 10:15 AM	Timestamp used in SLA scripts and flow conditions.
DenialReason	Multiple lines	No	Department not eligible for this sale	Populated by approver or admin when status = Denied.

List 3 — Approval Log

Purpose: A dedicated audit record for every approval decision. Created automatically by the Access Request Approval Flow when a manager responds to an approval email.

Relationships: RequestID → Access Requests; SaleID → Sale Master.

Column Name	Data Type	Req	Example Value	Purpose / Notes
LogID	Single line text	Yes	LOG-001	Unique identifier for the log record.
RequestID	Single line text	Yes	REQ-2026-0047	Links this log entry to the originating access request.
SaleID	Single line text	Yes	SALE-2026-001	Links to sale for filtering and reporting.
ApproverName	Single line text	Yes	Amanda Chen	Name of the person who made the approval decision.
ApproverEmail	Single line text	Yes	a.chen@luxemaison.com	Email of the approver — used in audit reporting.
Decision	Choice	Yes	Approved	Values: Approved; Denied. Set by the approval flow.
DecisionDate	Date and Time	Yes	5/10/2026 11:22 AM	Used to calculate approval turnaround time in SQL and Power BI.

Column Name	Data Type	Req	Example Value	Purpose / Notes
Comments	Multiple lines	No	Approved — eligible dept and location	Optional approver notes captured from approval email response.

List 4 — Issue Tracker

Purpose: All operational issues logged during sale setup, the active sale window, or post-sale cleanup. Priority = High triggers an immediate alert to the admin team via Flow 4.

Relationships: SaleID → Sale Master.

Column Name	Data Type	Req	Example Value	Purpose / Notes
IssueID	Single line text	Yes	ISS-001	Unique identifier. Referenced in notifications and escalation emails.
SaleID	Lookup / text	Yes	SALE-2026-001	Links issue to specific sale.
IssueCategory	Choice	Yes	Access — Employee	Values: Access — Employee; Access — Admin; Workflow Error; App Bug; Notification; Other.
Priority	Choice	Yes	High	Values: High; Medium; Low. High triggers immediate Flow 4 alert.
Status	Choice	Yes	Open	Values: Open; In Progress; Resolved; Closed. Used in all issue reports.
Description	Multiple lines	Yes	Approved employee cannot log in	Free text description of the issue.
ReportedBy	Single line text	Yes	Sofia Marchetti	Person who logged the issue.
ReportedDate	Date and Time	Yes	5/15/2026 10:30 AM	Timestamp used in resolution time calculations (SQL Scripts 07, 09).
AssignedTo	Single line text	No	IT Support	Person or team responsible for resolving the issue.
ResolvedDate	Date and Time	No	5/15/2026 12:45 PM	Populated when status changes to Resolved. Used in resolution time calculations.
IsEscalated	Yes/No	No	Yes	Flag for escalated issues. Used in Power BI Issues and Support page.
ResolutionNotes	Multiple lines	No	Resolved — recreated access link	Free text notes on how the issue was resolved.

List 5 — Notification Log

Purpose: Audit record of all automated notification emails sent by Power Automate flows. Used to verify delivery, troubleshoot missing notifications, and provide a communication audit trail.

Relationships: SaleID → Sale Master.

Column Name	Data Type	Req	Example Value	Purpose / Notes
NotifID	Single line text	Yes	NOTIF-001	Unique identifier for the notification record.
SaleID	Single line text	Yes	SALE-2026-001	Links notification to specific sale.

Column Name	Data Type	Req	Example Value	Purpose / Notes
Recipient	Single line text	Yes	s.marchetti@luxemaison.com	Email address of the notification recipient.
NotifType	Choice	Yes	Access Approved	Values: Access Approved; Access Denied; Launch Reminder; High Priority Issue; Status Update; Cleanup Reminder.
SentDate	Date and Time	Yes	5/10/2026 11:25 AM	Timestamp of when the notification was sent by the flow.
DeliveryStatus	Choice	Yes	Delivered	Values: Delivered; Failed. Updated based on flow run outcome.
FlowName	Single line text	No	Access Request Approval Flow	Name of the flow that sent the notification — for troubleshooting.

List 6 — Change History

Purpose: Comprehensive audit log of all significant configuration changes. Power Automate writes entries when key status fields change across Sale Master and Employee Access Requests.

Relationships: RecordID maps to SaleID or RequestID depending on the change type.

Column Name	Data Type	Req	Example Value	Purpose / Notes
ChangeID	Single line text	Yes	CHG-001	Unique identifier for the change record.
RecordID	Single line text	Yes	SALE-2026-001	ID of the record that was changed (SaleID or RequestID).
RecordType	Choice	Yes	Sale	Values: Sale; Access Request. Indicates which list the change occurred in.
ChangedBy	Single line text	Yes	Amanda Chen	Person who made the change.
ChangedDate	Date and Time	Yes	5/14/2026 3:30 PM	Timestamp of the change.
FieldChanged	Single line text	Yes	SaleStatus	Name of the field that was updated.
OldValue	Single line text	No	Readiness Review	Previous value before the change.
NewValue	Single line text	Yes	Active	New value after the change.
ChangeNote	Multiple lines	No	Launch authorised by ops manager	Optional note about the reason for the change.

List 7 — Sale Readiness Checklist

Purpose: Structured pre-launch checklist. Each record is one checklist item for one sale. A sale is considered ready to launch when all IsComplete values for that SaleID are true (or Yes). The Authorize Launch button in the app checks this condition before enabling.

Relationships: SaleID → Sale Master.

Column Name	Data Type	Req	Example Value	Purpose / Notes
ChecklistID	Single line text	Yes	CHK-001	Unique identifier for the checklist item.
SaleID	Lookup / text	Yes	SALE-2026-001	Links checklist item to specific sale.

Column Name	Data Type	Req	Example Value	Purpose / Notes
CheckItemName	Single line text	Yes	Sale setup fields validated	Name of the checklist item displayed in the app.
ItemDescription	Multiple lines	No	Verify all required sale fields are populated	Optional description of what this item requires.
AssignedTo	Single line text	No	Amanda Chen	Person responsible for completing this item.
IsComplete	Yes/No	Yes	Yes	Completion flag. Power Fx counts IsComplete = true to calculate readiness. Power BI may import this as TRUE/FALSE, Yes/No, or 1/0 — verify before writing DAX.
CompletedDate	Date and Time	No	5/14/2026	Populated when IsComplete is set to Yes.
CompletedBy	Single line text	No	Amanda Chen	Person who marked the item complete.

List 8 — User Role Reference

Purpose: Role assignment table used by Power Apps to determine navigation and feature visibility. The app reads this list on startup using the current user's email and stores the result in varCurrentUser.

Relationships: Standalone — no active list relationship. Linked via UserEmail at app load.

Column Name	Data Type	Req	Example Value	Purpose / Notes
UserEmail	Single line text	Yes	a.chen@luxemaison.com	Exact Microsoft 365 email address. Must match the value returned by Office() in Power Fx.
UserDisplayName	Single line text	Yes	Amanda Chen	Display name shown in the app UI.
Role	Choice	Yes	Operations Admin	Values: Operations Admin; Manager; Employee; Read-Only. Controls all navigation and visibility logic in the app.
Department	Choice	No	Operations	Used for filtering and reporting — not required for role lookup.
Location	Choice	No	New York Flagship	Used for filtering eligibility — not required for role lookup.
IsActive	Yes/No	No	Yes	Optional flag to deactivate a user without deleting the record.

SharePoint Permissions and Build Notes

App-Level vs. True Security

The User Role Reference list controls what users see inside the Power App — this is app-level visibility control only. It does not prevent a user from accessing SharePoint lists directly through a browser or the SharePoint API.

For a production deployment, app-level navigation control must be combined with:

- Power Apps sharing — only share the app with users who should have access
- SharePoint site permissions — configure Read, Contribute, or Edit at the site level
- List-level permissions — restrict sensitive lists (Approval Log, Change History) to admin groups only
- Microsoft 365 groups — use groups to manage large populations consistently

Choice Field Values Reference

All Choice field values across the system are listed below. These must be configured exactly as shown — Power Fx formulas, Power Automate conditions, and SQL scripts all reference these exact strings.

List	Column	Choice Values
Sale Master	SaleStatus	Draft Open for Requests Readiness Review Active Closing Closed Archived
Access Requests	RequestStatus	Pending Approved Denied Pending Revocation Revoked
Access Requests	EmployeeDepartment	Retail Operations Finance HR Visual Merchandising IT Operations Other
Access Requests	EmployeeLocation	New York Flagship Los Angeles Chicago Miami London Other
Approval Log	Decision	Approved Denied
Issue Tracker	IssueCategory	Access — Employee Access — Admin Workflow Error App Bug Notification Other
Issue Tracker	Priority	High Medium Low
Issue Tracker	Status	Open In Progress Resolved Closed
Notification Log	NotifType	Access Approved Access Denied Launch Reminder High Priority Issue Status Update Cleanup Reminder
Notification Log	DeliveryStatus	Delivered Failed
Change History	RecordType	Sale Access Request
User Role Reference	Role	Operations Admin Manager Employee Read-Only

Build tip: Create all Choice fields first, before adding any data. If you add a new choice value after records exist, Power Automate conditions and Power Fx If() expressions that reference the field may need to be updated. Agree on the full value list before building.